Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **full-stack AI products**, **Python backend APIs**, and **React/TypeScript experiences** with strong ownership in reliability, observability, and production delivery.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving reliability, observability, and secure service behavior in production.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **production stability** and issue triage across distributed systems used by product and research teams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, shipping polished product experiences from prototype to production while maintaining secure API boundaries and high-quality release standards.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through automated checks, maintainable tests, and rapid iteration with cross-functional stakeholders.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling reproducible behavior analysis and scalable data workflows that accelerated research-to-product iteration in AI-enabled development environments.
- Produced technical summaries and debugging diagnostics that translated findings into implementation plans, improving **cross-functional collaboration** and reducing ambiguity across engineering, research, and product execution cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across distributed services.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** via automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving service reliability and user-facing responsiveness in high-throughput collaborative operations.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through scripted validations and robust rollback mechanisms.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
Frontend: React, TypeScript, JavaScript, User Experience Prototyping, Product Iteration
Backend & APIs: Flask, FastAPI, REST APIs, Microservices, Distributed Services, Data Pipelines
Cloud & Reliability: AWS (S3, Glue, ECS), Docker, Kubernetes, Azure, Linux/UNIX, Observability
Engineering Practices: CI/CD, Unit/Integration Testing, Security-Aware Development, Git, Jenkins, Agile/Scrum

Projects

PiXMod | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built an AI visual editing platform using **SAM** segmentation and **Stable Diffusion** generation, processing **1K+ assets** while improving end-to-end product quality, responsiveness, and metadata-driven workflow reliability.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a semantic retrieval platform over **50K+ documents** using **FAISS** and vector search, delivering context-aware responses and scalable APIs for assistant-like, AI-native product experiences.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.